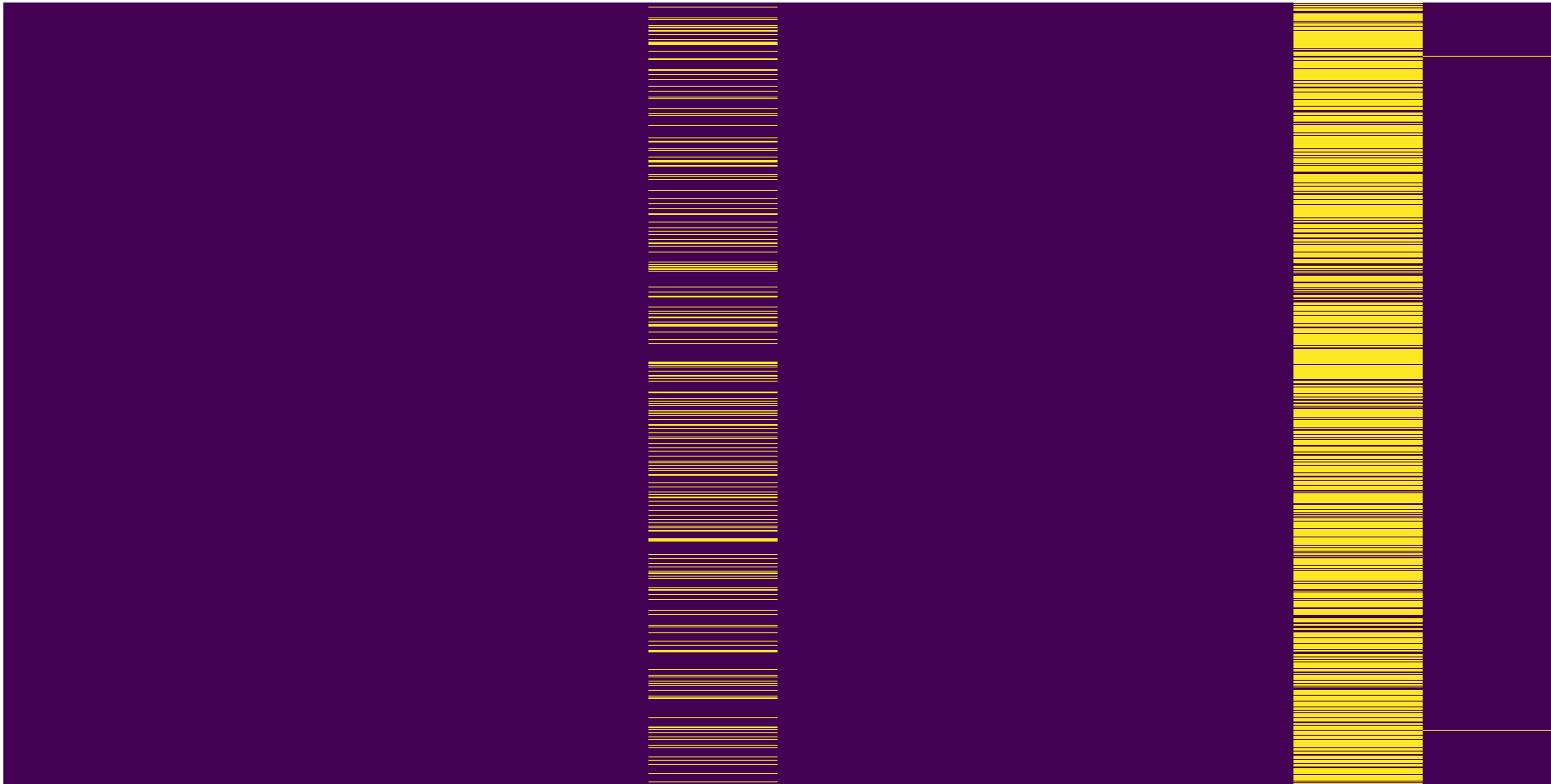


Machine Learning Lifecycle Mapping (Titanic Task)

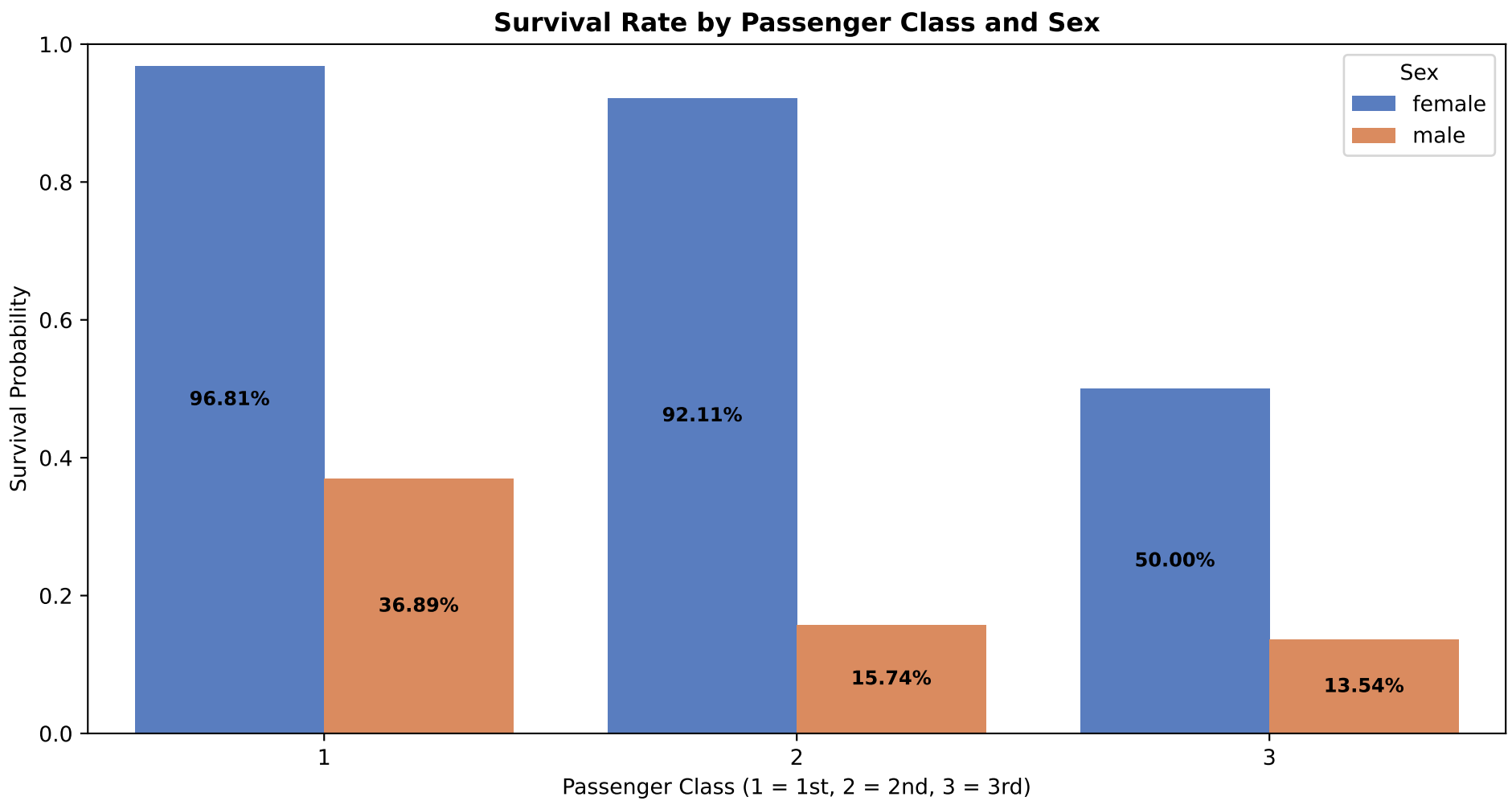
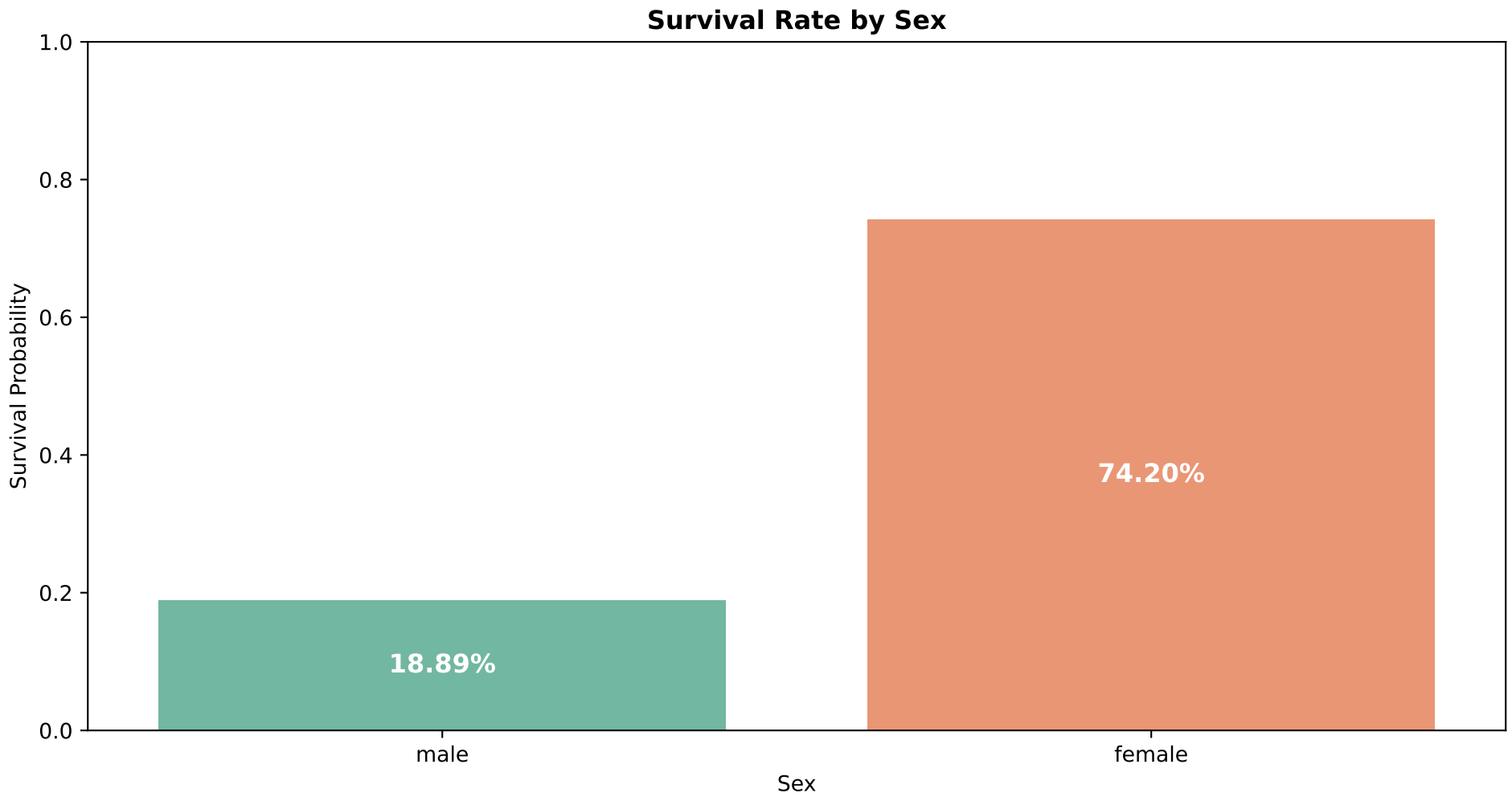
Phase	Step	Description & Implementation
Phase 1	Problem Definition	Binary Classification: Predict survival (0 = No, 1 = Yes)
Phase 2	Data Ingestion	Acquire passenger manifests with features (PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked)
Phase 3	Data Cleaning	Handle nulls in Age (median), Embarked (most frequent)
Phase 4	EDA & Exploration	Analyze distributions, class/gender bias, correlations
Phase 5	Feature Engineering	Extract Age Groups, Family Size, One-Hot Encode Cabin
Phase 6	Modeling & Eval	Train classifiers (Logistic, Trees), evaluate ROC AUC
Phase 7	Deployment	Package best pipeline as REST API service for predictions

Raw Dataset Missing Values Heatmap (Before Cleaning)

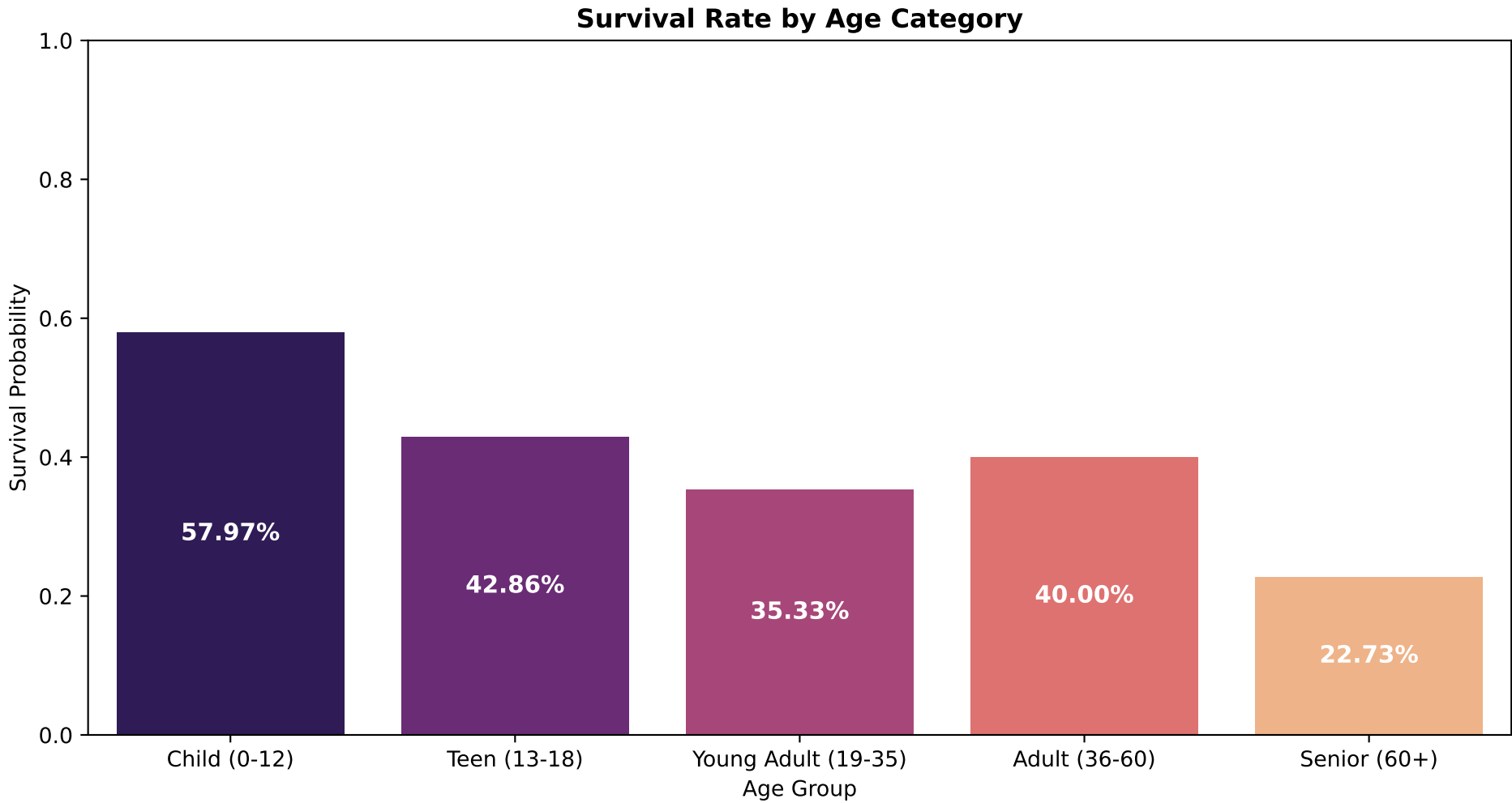
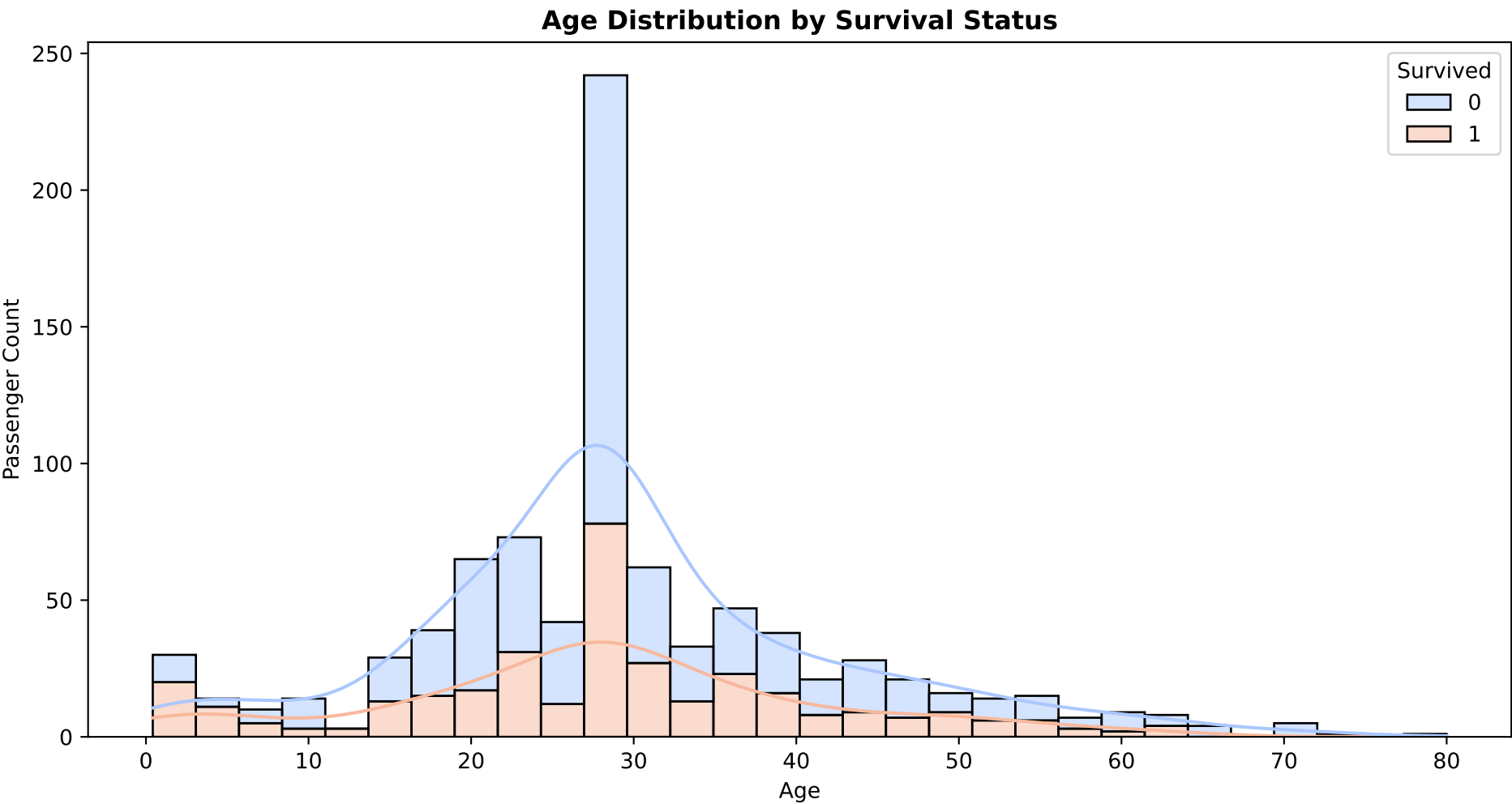


PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

Survival Analysis: Demographics & Socioeconomic Status



Survival Analysis: Age Demographics



Correlation Matrix & Fare Exploration

